



Organizational Training (OT)

OT Overview

Required PA Information

Intent

Develops the skills and knowledge of personnel so they perform their roles efficiently and effectively.

Value

Enhances individuals' skills and knowledge to improve organizational work performance.

Additional Required PA Information

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Explanatory PA Information

Practice Summary

Level 1

OT 1.1 Train people.

Level 2

OT 2.1 Identify training needs.

OT 2.2 Train personnel and keep records.

Level 3

OT 3.1 Develop and keep updated the organization's strategic and short-term training needs.

OT 3.2 Coordinate training needs and delivery between the projects and the organization.

OT 3.3 Develop, keep updated, and follow organizational strategic and short-term training plans.

OT 3.4 Develop, keep updated, and use a training capability to address organizational training needs.

OT 3.5 Assess and report the effectiveness of the organization's training program.

OT 3.6 Record, keep updated, and use the set of organizational training records.

Additional PA Explanatory Information

The training program supports the organization's business objectives, and training needs that are common across projects and support groups. The organization coordinates with projects and support groups to determine responsibility for providing training.

Provide training to enhance skills and knowledge including:

- Technical skills related to:
 - Using equipment, tools, materials, and data
 - Processes required by the work or the organization
- Organizational skills related to:
 - Role and responsibilities
 - General operating principles and methods
 - Behavior within the organization
- Contextual skills related to:
 - Self-management
 - Communication
 - Interpersonal abilities needed to successfully perform work

Deliver training either informally, e.g., brown bags, task/job walk-throughs; or formally, e.g., classroom, electronic-based training, workshops, guided self-study, formalized on-the-job training.

[Figure OT-1: Elements in an Organizational Training Program](#) shows the primary elements involved in an organizational training program.

Figure OT-1: Elements in an Organizational Training Program



An organizational training program involves:

- Determining and identifying the organizational training needs
- Planning for training
- Developing a training capability and keeping it updated to:
 - Identify how training will be obtained or developed
 - Develop, design, and keep updated training materials, schedules, records, and related data
 - Develop mechanisms for measuring the effectiveness of the training and the training program
 - Identify personnel with training delivery skills and subject matter expertise to deliver training
- Providing training
- Keeping records of completed training
- Maintaining and using the organizational set of training records
- Assessing training effectiveness against organizational training needs and communicate results

Related Practice Areas

Process Asset Development (PAD)

Workforce Empowerment (WE)

Context Specific

Data

Context Tag:	CMMI-DATA
Context:	Use processes to incorporate data management best practices as an integral part of the solution.

Consider requirements, processes, and risks when identifying training needs relevant to managing data. Ensure training needs are aligned to roles and responsibilities involved in managing data.

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Consider affected stakeholders of the service system for organizational training activities. Affected stakeholders include customers, end users, provider personnel, senior management, external suppliers, and anyone else who is engaged with the service system. Develop, keep updated, and follow training plans that include the appropriate forms of training and communications to affected stakeholders. Training for affected stakeholders can vary greatly based on the complexity of the service system, scope of transition changes to the service system, and the knowledge and skills of the stakeholders.

Level 1

OT 1.1

Required Practice Information

Practice Statement

Train people.

Value

Increases likelihood of meeting objectives by ensuring individuals have needed skills and knowledge.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

This section left blank for future content.

Example Activities

Example Activities	Further Explanation
Identify people who will receive the training.	
Schedule training.	
Deliver the training.	

Example Work Products

Example Work Products	Further Explanation
Completed training	

Level 2

OT 2.1

Required Practice Information

Practice Statement

Identify training needs.

Value

Reduces costs by providing training needed to perform the work.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Identify the skills and knowledge needed for each role. Compare this to the skills and knowledge of the individuals in those roles to determine what training is needed.

Example Activities

Example Activities	Further Explanation
Identify skills and knowledge for each role.	
Record and keep updated skills and knowledge of individuals.	
Perform a gap analysis to determine training needs.	
Record and communicate training needs.	

Example Work Products

Example Work Products	Further Explanation
List of training needs	List of needs should include: • Skills and knowledge by role • Skills and knowledge for individuals • Gaps for needed training • Technical skills • Soft skills ○ Communication ○ Emotional intelligence ○ Teamwork ○ Problem solving

Example Work Products	Further Explanation
	○ Active listening ○ Time management ○ Meeting management ○ Interpersonal communication and dynamics ○ Multicultural sensitivity, diversity, inclusion, equity, and other related skills ● Managerial skills ○ Coaching ○ Mentorship ○ Decision making ○ Problem resolution ○ Conflict resolution ○ Negotiation ○ Stress management ● Team building

Related Practice Areas

Planning (PLAN)

Context Specific

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Consider safety requirements, awareness, and risks when identifying training needs. Ensure safety-related roles and responsibilities are aligned to training needs.

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Consider security requirements, awareness, and risks when identifying training needs. Ensure security-related roles and responsibilities are aligned to training needs.

OT 2.2

Required Practice Information

Practice Statement

Train personnel and keep records.

Value

Avoids training people who already have the needed knowledge and skills. and verifies that people get the training needed to perform their work.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

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Example Activities

Example Activities	Further Explanation
Deliver training based on identified training needs.	
Maintain training records.	

Example Work Products

Example Work Products	Further Explanation
Records of delivered training	Include: • Training name • Description of delivered training • Date completed • Student names • Instructor names • Pass/fail indicator

Level 3

OT 3.1

Required Practice Information

Practice Statement

Develop and keep updated the organization's strategic and short-term training needs.

Value

Maximizes the likelihood of meeting objectives by ensuring that the organization has skilled individuals now and in the future.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Strategic training needs address long-term business objectives to build capability by:

- Filling significant knowledge gaps
- Keeping current with new and emerging technologies
- Adapting to changing business needs

Strategic training considerations typically cover a span of time into the future beyond the short-term training activities. Strategic training is typically linked to organizational strategic needs and objectives.

Examples of sources of strategic training needs include:

- The organization's business plan
- The organization's standard processes
- The organization's process improvement plan
- Enterprise-level initiatives
- Skill assessments showing common long-term needs
- Industry trends

The primary difference between strategic and short-term needs is how quickly that competency is needed.

Short-term training needs address immediate business objectives to build capability by:

- Filling current knowledge gaps
- Responding to an event driven need
- Introducing new, currently needed technologies

When developing strategic and short-term training needs, consider:

- The organization's business plan
- The organization's standard processes
- The organization's process improvement plan
- Background of the target population of training participants
- Need for cross-discipline training
- Need for training in organizational processes
- Need for training in disciplines related to support functions
- Needs arising from competency development plans
- Need to keep personnel competencies and qualifications of personnel updated for performing current and future work

Example Activities

Example Activities	Further Explanation
Determine the roles and skills needed to perform the organization's set of standard processes and the work to be performed.	
Determine training needs.	Analyze and prioritize the strategic and short-term training needs.
Record and keep updated the prioritized strategic and short-term organization training needs.	For example, training needed to: • Perform roles in the organization's set of standard processes • Maintain the safe, secure, and continued operation of the business
Review the organization's training needs periodically and on an event driven basis and update the needs if required.	

Example Work Products

Example Work Products	Further Explanation
Training needs	

Related Practice Areas

Workforce Empowerment (WE)

OT 3.2

Required Practice Information

Practice Statement

Coordinate training needs and delivery between the projects and the organization.

Value

Ensures efficient and effective allocation of training resources.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Organizational training addresses training needs and requirements that are common across projects. The organization's training personnel may also provide or support the additional training needs of the projects and organization based on the organization's priorities and training resources.

When scheduling training, courses should consider opportunities for personnel to use the new skills in a timely manner.

Example Activities

Example Activities	Further Explanation
Analyze the training needs identified by work and support groups.	Identify common training needs that can be addressed across the entire organization. Consider: • technical training • soft skills training • competency gap identification • competency analysis and development • grievance redressal • cultural awareness Use analysis results to predict future training needs at the work and support group level.
Coordinate with projects and support groups to determine how training needs will be addressed.	
Record responsibilities among the organization, project, and support groups for delivering training.	
Coordinate opportunities to reinforce new skills and knowledge from training activities.	This is necessary to establish new skills and knowledge as habitual and persistent.

Example Work Products

Example Work Products	Further Explanation
Allocated training needs	May include: • Organization common needs • Project needs • Support group needs
Training delivery responsibilities	Allocated to the: • Organization • Projects • Support groups

Context Specific

People

Context Tag:	CMMI-PPL
Context:	Use processes to align workforce performance and business objectives.

Conduct personal development discussions, e.g., experience, competencies, career development, with each individual. Create personalized individual development plans, based on training needs and an assessment of individual skills as compared to the organization's workforce competency descriptions or the needs of the project. Include monitoring parameters within the development plan.

OT 3.3

Required Practice Information

Practice Statement

Develop, keep updated, and follow organizational strategic and short-term training plans.

Value

Increases effectiveness and efficiency of task performance.

Additional Required Information

Incorporate competency development opportunities within training plans.

If Safety is included in the selected view: Ensure safety and safety awareness training is incorporated within the organizational strategic and short-term training plans. Ensure content includes roles and responsibilities involved with safety, schedule of activities, and approach to training delivery.

If Security is included in the selected view: Ensure security and security awareness training is incorporated within the organizational strategic and short-term training plans. Ensure content includes roles and responsibilities involved with security, schedule of activities, and approach to training delivery.

Explanatory Practice Information

Additional Explanatory Information

The strategic training plan describes how the organization will deliver training to meet the recorded long-term needs. The organizational training short-term plan addresses near-term training implementation. Adjust the plans periodically in response to changes and evaluations of effectiveness.

Example Activities

Example Activities	Further Explanation
Develop the content of the organizational training strategic plan.	Plan should address: • Strategic training needs • Approach to delivering training • Approach to competency development • Prioritization • Timeframes • Training effectiveness evaluations • Resources • Mentorship programs • Coaching options and parameters
Develop the content of the organizational training short-term plan.	Organizational training short-term plans typically contain: • Training needs • Training topics • Schedules and their dependencies • Methods used for training • Requirements and quality standards for training materials • Training tasks, roles, and responsibilities • Ensuring training effectiveness • Required resources such as: ○ Tools ○ Facilities ○ Environments ○ People
Review plans, determine commitments, and communicate the review results to affected stakeholders.	
Revise the plans and commitments as needed.	

Example Work Products

Example Work Products	Further Explanation
Organizational training strategic plan	
Organizational training short-term plan	
Recorded commitments	

Related Practice Areas

Planning (PLAN)

OT 3.4

Required Practice Information

Practice Statement

Develop, keep updated, and use a training capability to address organizational training needs.

Value

Ensures personnel have the knowledge, skills, and abilities to perform their work efficiently and effectively.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Provide the training to meet the needs of the organization, projects, and support groups.

Example Activities

Example Activities	Further Explanation
Select approaches to satisfy organizational training needs.	To select an approach, consider how to provide skills and knowledge in the most efficient and effective way. Factors that can affect the selection of training approaches, include: • Learner knowledge • Constraints • Costs • Schedule • Work environment Examples of training approaches include:

Example Activities	Further Explanation
	• Classroom training • Computer or technology aided instruction, e.g., tutorials, simulations, webinars • Guided self-study • Formal apprenticeship • Mentorship • Coaching • Facilitated videos • Chalk talks • Brown bag lunch seminars • Structured on-the-job training
Decide whether to develop training internally or to acquire it externally.	Decision criteria may include: • Preparation time • Cost benefit analysis • Availability of in-house expertise • Availability of external training • Training effectiveness data Examples of external sources of training may include: • Customer-provided training • Commercially available training courses • Academic programs • Professional conferences • Seminars
Develop or obtain training materials.	Examples of training materials include: • Curriculum • Syllabus • Handouts • Books • Exercise instructions • Videos and interactive media • Virtual environments • Simulations or labs • Mobile applications
Identify, train, or hire qualified instructors, instructional designers, or mentors.	For those who develop and deliver internal training, consider: • Subject matter expertise • Relevant training skills and experience • Experience in instructional design • Mentoring skills
Describe the training in the organization's training curriculum.	Examples of information provided in training descriptions include: • Topics covered in the training • Intended audience

Example Activities	Further Explanation
	• Prerequisites and participant preparation • Training objectives • Length of the training • Lesson plans • Completion criteria for the course • Criteria for granting training waivers
Review training periodically and on an event-driven basis.	For example, review for: • Compliance with defined standards ○ Instructional design standards ○ Content standards • Revisions needed in the standards • Effectiveness
Revise training materials as needed.	
Identify and make available resources for delivering the training.	
Update the training and development program.	
Keep training records updated.	
Communicate availability of training.	

Example Work Products

Example Work Products	Further Explanation
Training materials and supporting artifacts	
List of courses	
Training records	
List of instructors	May include certifications and experience.
Instruction design standards	
Training facilities and resources	

Related Practice Areas

Decision Analysis and Resolution (DAR)

Context Specific

Data

Context Tag:	CMMI-DATA
Context:	Use processes to incorporate data management best practices as an integral part of the solution.

Consider training needs for individuals relative to managing data. Establish mechanisms to ensure all individuals are aware of the principles, processes, regulations, and guidelines, e.g., how to access and use the data glossary for managing data.

Strong data practices often require Subject Matter Experts (SMEs) for execution. These experts are trained and, where appropriate, certified in their specific discipline. Career paths and professional growth plans should be established to ensure that personnel have the means to improve and sharpen their skills for the benefit of the organization. Skilled personnel resources are valuable assets of the organization, and their skills and competencies should be identified, formally recognized, and leveraged across the organization. This helps to guide others with less training, supports consistency of practices across the organization, strengthens legal and regulatory compliance regarding an organization's data, and facilitates persistent and habitual performance of sound activities for managing data.

DevSecOps

Context Tag:	DevSecOps
Context:	DevSecOps is a mindset, a culture, and set practices that fosters close cooperation between development, operations, and security to plan, develop, test, deploy, release, and keep updated a secure solution.

Leveraging the DevSecOps concepts, there is a focus to automate as much of the process as possible. Organizations with DevSecOps teams need to tightly couple their development process and technical training to ensure the teams have adequate skills needed to build and maintain an automated Continuous Integration / Continuous Delivery (CI/CD) process using the tools provided by the organization efficiently and effectively. Technical training on the organization's tool suite needs to be scheduled on a recurring basis to account for personnel turnover and to stay current with updates to tools and environments. Tool guidelines may be kept updated at the team level to support operational tool use. Mentoring within the team is also an effective training approach. The organization should also focus on improving soft skills to improve collaboration between the operations and development teams.

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Identify topics that require safety awareness training for a broader audience versus more technical, in-depth safety training. Determine the most effective delivery approach for each type of training, e.g., internal versus external and classroom versus computer-based training (CBT). Establish mechanisms to ensure safety training and related materials remain consistent with industry, e.g., consistent with laws, considerate of technology changes, and organization information. Ensure content of safety training materials incorporates safety topics and attributes, recognizing and addressing issues, emergency safety procedures, and scenarios.

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Identify topics that require security awareness training for a broader audience, versus more technical, in-depth security training. Determine the most effective delivery approach for each type of training, e.g., internal versus external and classroom versus computer-based training (CBT). Establish mechanisms to ensure security training and related materials remain consistent with industry, e.g., consistent with industry trends, considerate of technology changes, and organization information. Ensure content of security training materials incorporates all aspects of security relevant to the organization, e.g., physical security, standards and processes, access control, customer data, solution functionality, and security considerations of third-party systems.

OT 3.5

Required Practice Information

Practice Statement

Assess and report the effectiveness of the organization's training program.

Value

Keeps the training program relevant and valuable to the business.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Develop a process to determine how effectively training:

- Improves the capability of people to perform their work
- Enables achieving performance improvement goals
- Meets the organization's needs and objectives
- Meets the organization's standards for the quality of the training

Effective assessment of training includes:

- Training assessments to determine before and after levels of competence
- Measurement scales of training effectiveness

To ensure that the information is not used for positive or negative individual incentives, individuals' training effectiveness assessments should not be reported to their managers.

The impact and results of completed training activities are analyzed to determine if adjustments are needed. The results of personnel development activities are recorded, monitored, and used as inputs into future training needs analyses.

Example Activities

Example Activities	Further Explanation
Assess the effectiveness of each training course.	Examples of methods used to assess training course effectiveness may include: • Testing in the training context, e.g., meeting learner objectives • Assessment mechanisms embedded in courseware • Evaluations of instructor effectiveness • Training assessments involving training participants and their managers to determine if the training delivered has helped the projects
Assess the effectiveness of the training programs.	Examples of methods used to assess training program effectiveness may include: • Analysis of the improvement in performance of individuals to determine effectiveness of the: ○ Training provider ○ Instructors ○ Course materials ○ Overall program • Process noncompliances; these can be considered as an input and may indicate an issue with the training program • Surveys on the training program effectiveness • Industry standards for benchmarking training effectiveness against behavioral and performance improvements goals
Report the assessment results to stakeholders.	

Example Work Products

Example Work Products	Further Explanation
Training effectiveness surveys	May include both individual course surveys and overall program surveys
Training program assessments	Assessments may include instructor observations.

Example Work Products	Further Explanation
Training program analysis results	
Instructor evaluation forms	May include trend and pattern analysis across multiple courses
Training examinations	
Training measures	May include: • Training assessment results • Achievement of benchmarking goals

OT 3.6

Required Practice Information

Practice Statement

Record, keep updated, and use the set of organizational training records.

Value

Records are essential in determining how well the training program supports the achievement of business and performance goals.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Ensure that training records are kept for individuals, projects, and support groups, and organizational training.

Example Activities

Example Activities	Further Explanation
Keep and use records of all participants.	
Keep and use records of all personnel who are waived from training.	Record the rationale.
Keep and use records of the training course and program effectiveness.	May include: • Course evaluations and other feedback mechanisms and information • Improvement of individual performance • Contextual information of provided training: ○ Instructor ○ Course name

Example Activities	Further Explanation
	○ Provider ○ Date delivered ○ Students • Course feedback
Make training records available to the appropriate people for consideration in assignments.	Training records may be part of a skills matrix summarizing the experience and education of people.

Example Work Products

Example Work Products	Further Explanation
Training records and reports	

